

# **Zoobenthic Community Indicators of Sediment Contamination**

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# Chapter 1

## Introduction

- 1.1 01 pollution assessment
- 1.2 02 Taxa Assemblage in Reference Sites
- 1.3 03 Env-Driven Taxa clusters (LDA)

Table 1.1: Table1 Pca Loadings

| var                | PC1     | PC2     | PC3     | PC4     | PC5     | PC6     |
|--------------------|---------|---------|---------|---------|---------|---------|
| %OC                | 0.2398  | 0.0102  | -0.0619 | 0.1267  | 0.0017  | -0.0274 |
| 1234-TCB           | 0.0615  | -0.0146 | 0.0493  | 0.2244  | 0.1144  | 0.7924  |
| 1245-TCB           | 0.1046  | 0.3572  | 0.0924  | -0.2376 | 0.0223  | -0.0311 |
| Al                 | 0.2492  | -0.0293 | -0.189  | 0.0399  | 0.1703  | -0.1193 |
| As                 | 0.1424  | -0.1519 | -0.0708 | -0.2066 | 0.0894  | -0.0452 |
| Bi                 | -0.0995 | 0.2882  | -0.0307 | 0.3978  | -0.0163 | -0.1473 |
| Ca                 | 0.1489  | 0.241   | -0.0924 | 0.2842  | 0.0188  | 0.0651  |
| Cd                 | 0.1345  | 0.1453  | 0.2956  | 0.2916  | -0.2262 | 0.0348  |
| Co                 | 0.253   | -0.028  | -0.1276 | -0.0369 | 0.1393  | -0.0863 |
| Cr                 | 0.2551  | -0.1152 | 0.0486  | -0.0123 | -0.1472 | 0.0105  |
| Cu                 | 0.229   | -0.1514 | 0.0189  | -0.0401 | -0.2513 | -0.0427 |
| Fe                 | 0.2551  | -0.0512 | -0.0914 | -0.1007 | 0.1471  | 0.1437  |
| HCB                | 0.0663  | 0.3687  | 0.1068  | -0.2045 | 0.1815  | 0.1606  |
| Heptachlor Epoxide | -0.0389 | 0.033   | 0.2035  | 0.1062  | 0.5047  | 0.1616  |
| Hg                 | 0.1189  | 0.1751  | 0.3562  | 0.0092  | -0.0668 | -0.1937 |
| K                  | 0.2391  | 0.0428  | -0.1782 | 0.1384  | 0.174   | -0.1253 |
| Mg                 | 0.1706  | 0.216   | -0.1355 | 0.2701  | 0.0167  | -0.0569 |
| Mn                 | 0.2498  | 0.018   | -0.1643 | 0.1023  | 0.0545  | -0.0125 |
| Na                 | 0.2224  | 0.1312  | -0.1279 | 0.0126  | -0.0682 | -0.121  |
| Ni                 | 0.2736  | -0.0502 | -0.0128 | -0.0007 | -0.056  | -0.0151 |
| OCS                | 0.0626  | 0.3425  | 0.0228  | -0.3233 | -0.0102 | -0.0603 |
| Pb                 | 0.2124  | -0.0839 | 0.233   | -0.0629 | -0.2868 | 0.0853  |
| QCB                | 0.0745  | 0.3684  | 0.0934  | -0.2719 | 0.1858  | 0.0257  |
| Sb                 | -0.1723 | 0.227   | 0.0154  | 0.3304  | -0.0858 | -0.1821 |
| V                  | 0.2438  | -0.0704 | -0.1666 | 0.037   | 0.1685  | -0.0154 |
| Zn                 | 0.2515  | -0.0491 | 0.0976  | 0.0636  | -0.1976 | 0.1026  |
| mirex              | 0.0345  | -0.111  | 0.3583  | 0.1752  | 0.1934  | -0.1004 |
| p,p'-DDD           | 0.1153  | -0.1551 | 0.3442  | 0.0049  | 0.1645  | -0.2055 |
| p,p'-DDE           | 0.0656  | -0.2257 | 0.3157  | 0.086   | 0.3694  | -0.2138 |
| total PCB          | 0.1727  | 0.0612  | 0.3152  | -0.0677 | -0.2141 | 0.1419  |

Table 1.2: Table2 Rda Axes Summary

| Axis | Eigenvalue | Expl. (%) | Cumul. (%) | F-stat | p-value |
|------|------------|-----------|------------|--------|---------|
| RDA1 | 0.0208     | 36.0749   | 36.0749    | 2.6938 | 0.296   |
| RDA2 | 0.0132     | 22.88     | 58.9549    | 1.7085 | 0.876   |
| RDA3 | 0.0088     | 15.1661   | 74.1211    | 1.1325 | 0.997   |
| RDA4 | 0.0073     | 12.6117   | 86.7328    | 0.9418 | 1       |
| RDA5 | 0.0044     | 7.6547    | 94.3875    | 0.5716 | 1       |
| RDA6 | 0.0032     | 5.6125    | 100        | 0.4191 | 1       |

Table 1.3: Table3 Rda Predictor Terms

| Variable | Inertia | F-stat | p-value | RDA1    | RDA2    |
|----------|---------|--------|---------|---------|---------|
| PC1      | 0.3345  | 1.0545 | 0.404   | 0.706   | -0.0236 |
| PC2      | 0.4795  | 1.5113 | 0.125   | -0.5998 | 0.2667  |
| PC3      | 0.3042  | 0.9589 | 0.477   | 0.0826  | -0.5524 |
| PC4      | 0.4403  | 1.3878 | 0.18    | -0.0164 | 0.3057  |
| PC5      | 0.451   | 1.4215 | 0.155   | -0.4908 | -0.2401 |
| PC6      | 0.2607  | 0.8216 | 0.588   | 0.4716  | -0.5281 |

Table 1.4: Table3 Rda Statistics

| Statistic               | Value   |
|-------------------------|---------|
| R <sup>2</sup>          | 0.1758  |
| Adjusted R <sup>2</sup> | 0.0346  |
| Total Inertia           | 13.4728 |
| Constrained Inertia     | 2.369   |
| Unconstrained Inertia   | 11.1037 |
| N Reference Sites       | 42      |

Table 1.5: Table0 Env Cluster Anova

| Variable                           | Cluster 0    | Cluster 1    | F-stat | p-value |
|------------------------------------|--------------|--------------|--------|---------|
| Measured Depth (m)                 | 4.24 ± 3.24  | 3.37 ± 2.20  | 1.19   | 0.281   |
| Velocity at bottom (m/sec) Imputed | 0.13 ± 0.14  | 0.12 ± 0.10  | 0.08   | 0.778   |
| Water DO Bottom (mg/L)             | 9.20 ± 1.26  | 9.63 ± 0.46  | 2.29   | 0.137   |
| Temperature (oC)                   | 20.70 ± 2.25 | 20.11 ± 0.96 | 1.34   | 0.252   |
| MPS (Phi)                          | 0.93 ± 1.22  | 1.44 ± 0.66  | 3.15   | 0.082.  |
| LOI (%)                            | 1.88 ± 1.19  | 1.50 ± 1.05  | 1.43   | 0.237   |
| Sample size (n)                    | 30           | 22           |        |         |

Table 1.6: Table1 Taxa Cluster Anova

| Variable          | Cluster 0   | Cluster 1   | F-stat | p-value   |
|-------------------|-------------|-------------|--------|-----------|
| Oligochaeta       | 4.32 ± 1.38 | 5.54 ± 0.65 | 29.56  | 0.0000*** |
| Chironomidae      | 3.70 ± 1.76 | 4.88 ± 1.05 | 25.25  | 0.0000*** |
| Nematoda          | 2.63 ± 1.79 | 2.35 ± 1.55 | 0.37   | 0.543     |
| Dreissena         | 2.66 ± 2.18 | 0.11 ± 0.35 | 36.19  | 0.0000*** |
| Amphipoda         | 1.70 ± 1.57 | 0.01 ± 0.04 | 34.54  | 0.0000*** |
| Sphaeriidae       | 1.00 ± 1.11 | 0.91 ± 1.45 | 0.05   | 0.817     |
| Gastropoda        | 1.44 ± 1.63 | 0.11 ± 0.29 | 17.25  | 0.0001*** |
| Caenis            | 0.88 ± 1.49 | 0.12 ± 0.28 | 3.66   | 0.061.    |
| Hexagenia         | 0.56 ± 1.11 | 0.51 ± 0.92 | 0.05   | 0.831     |
| Hydropsychidae    | 0.67 ± 1.38 | 0.33 ± 0.84 | 0.51   | 0.480     |
| Acari             | 0.39 ± 0.70 | 0.67 ± 0.93 | 3.13   | 0.083.    |
| Turbellaria       | 0.78 ± 1.10 | 0.08 ± 0.25 | 9.45   | 0.003**   |
| Other Trichoptera | 0.50 ± 1.09 | 0.37 ± 0.93 | 0.11   | 0.736     |
| Hydrozoa          | 0.58 ± 1.24 | 0.08 ± 0.18 | 1.89   | 0.175     |
| Hirudinea         | 0.21 ± 0.52 | 0.11 ± 0.51 | 0.66   | 0.419     |
| Ceratopogonidae   | 0.06 ± 0.35 | 0.02 ± 0.12 | 0      | 0.960     |
| Sample size (n)   | 30          | 22          |        |           |

Table 1.7: Table1 Rda Axes Summary

| Axis | Eigenvalue | Explained (%) | Cumulative (%) | F-statistic | p-value | Significance |
|------|------------|---------------|----------------|-------------|---------|--------------|
| RDA1 | 0.0344     | 52.9311       | 52.9311        | 6.2749      | 0.001   | **           |
| RDA2 | 0.0121     | 18.6191       | 71.5502        | 2.2072      | 0.551   | ns           |
| RDA3 | 0.0072     | 11.1062       | 82.6565        | 1.3166      | 0.986   | ns           |
| RDA4 | 0.0058     | 8.8746        | 91.5311        | 1.0521      | 0.998   | ns           |
| RDA5 | 0.0034     | 5.1669        | 96.698         | 0.6125      | 1       | ns           |
| RDA6 | 0.0021     | 3.302         | 100            | 0.3914      | 1       | ns           |

Table 1.8: Table2 Rda Terms Summary

| Environmental Variable        | Delta Inertia     | F-statistic       | p-value        | RDA1 Coef         | RDA2 Coef         |
|-------------------------------|-------------------|-------------------|----------------|-------------------|-------------------|
| MPS (Phi)                     | 0.8754            | 3.1356            | 0.003          | -0.887            | 0.0014            |
| Measured Depth (m)            | 0.5242            | 1.8774            | 0.028          | 0.6703            | -0.35             |
| <del>Velocity at bottom</del> | <del>0.3706</del> | <del>1.3272</del> | <del>0.2</del> | <del>0.5773</del> | <del>0.0589</del> |
| Temperature (oC)              | 0.31              | 1.1102            | 0.338          | -0.2675           | 0.7438            |
| Water DO Bottom (mg/L)        | 0.2511            | 0.8995            | 0.53           | -0.2128           | -0.4867           |
| LOI (%)                       | 0.2295            | 0.8218            | 0.611          | 0.231             | 0.508             |

Table 1.9: Table4 Lda Variable Importance

| Environmental Variable | Delta Wilks' Lambda | F-statistic | p-value | LD1 Coef |
|------------------------|---------------------|-------------|---------|----------|
| MPS (Phi)              | 0.0894              | 4.8716      | 0.0324  | 0.8826   |
| Temperature (oC)       | 0.0461              | 2.5148      | 0.1198  | -0.6173  |
| Velocity at bottom     | 0.0143              | 0.781       | 0.3815  | 0.344    |
| Measured Depth (m)     | 0.0108              | 0.5913      | 0.4459  | -0.2866  |
| LOI (%)                | 0.0084              | 0.456       | 0.503   | -0.2272  |
| Water DO Bottom (mg/L) | 0.0026              | 0.1424      | 0.7077  | 0.1347   |

Table 1.10: Table5 Lda Confusion Matrix

|                | Pred. Cluster 0 | Pred. Cluster 1 |
|----------------|-----------------|-----------------|
| True Cluster 0 | 24              | 6               |
| True Cluster 1 | 8               | 14              |

Table 1.11: Table6 Lda Classification Report (Note: Overall Accuracy = 0.7308; Number of sites = 52.0 )

|              | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Cluster 0    | 0.75      | 0.8    | 0.77     | 30      |
| Cluster 1    | 0.7       | 0.64   | 0.67     | 22      |
| Accuracy     | —         | —      | 0.73     | 52      |
| Macro avg    | 0.72      | 0.72   | 0.72     | 52      |
| Weighted avg | 0.73      | 0.73   | 0.73     | 52      |

Table 1.12: Table7 Mccv Confusion Matrix (Test Size: 20.0% per iteration and Note: Combined results from 1,000 cross-validation iterations)

|                | Pred. Cluster 0 | Pred. Cluster 1 |
|----------------|-----------------|-----------------|
| True Cluster 0 | 4245            | 1755            |
| True Cluster 1 | 2257            | 2743            |

Table 1.13: Table8 Mccv Classification Report (Note: Classification metrics shown as mean  $\pm$  standard deviation across 1,000 CV iterations )

|                 | Precision         | Recall            | F1-Score          | Support |
|-----------------|-------------------|-------------------|-------------------|---------|
| Cluster 0       | 0.661 $\pm$ 0.137 | 0.708 $\pm$ 0.202 | 0.669 $\pm$ 0.141 | 6       |
| Cluster 1       | 0.638 $\pm$ 0.205 | 0.549 $\pm$ 0.215 | 0.564 $\pm$ 0.172 | 5       |
| Weighted Avg    | 0.651 $\pm$ 0.151 | 0.635 $\pm$ 0.131 | 0.622 $\pm$ 0.137 | 11      |
| Mean accuracy   |                   |                   | 0.635 $\pm$ 0.131 |         |
| Median accuracy |                   |                   | 0.6364            |         |

Table 1.14: Table1 Species Pc Loadings

| Cluster | Taxon             | PC1     | PC2     | PC3     | PC4     | PC5     | PC6     | PC7     |
|---------|-------------------|---------|---------|---------|---------|---------|---------|---------|
| 0       | Dreissena         | 0.558   | 0.5214  | 0.1035  | -0.0131 | -0.2544 | -0.3619 | -0.1815 |
| 0       | Caenis            | 0.1021  | -0.4493 | 0.2344  | -0.1719 | -0.1692 | -0.2957 | 0.4786  |
| 0       | Hexagenia         | -0.1683 | -0.0344 | 0.1718  | -0.4159 | 0.5493  | -0.3267 | -0.1987 |
| 0       | Gastropoda        | 0.2443  | -0.2252 | 0.68    | -0.0233 | -0.0555 | 0.46    | -0.1377 |
| 0       | Nematoda          | -0.3449 | 0.5399  | 0.2238  | -0.2883 | 0.0076  | 0.0975  | 0.2117  |
| 0       | Sphaeriidae       | 0.0226  | -0.0604 | 0.2869  | 0.7051  | 0.2459  | -0.354  | 0.0002  |
| 0       | Amphipoda         | 0.3533  | 0.1152  | -0.1715 | 0.0027  | 0.1557  | -0.0829 | 0.5791  |
| 0       | Other Trichoptera | 0.1666  | -0.208  | -0.1909 | -0.0855 | 0.1217  | -0.183  | -0.4273 |
| 0       | Hydropsychidae    | 0.2933  | -0.214  | -0.3218 | -0.2355 | -0.0176 | 0.0447  | -0.1525 |
| 0       | Turbellaria       | 0.0374  | 0.1069  | -0.1183 | 0.1717  | 0.5851  | 0.1372  | 0.118   |
| 0       | Chironomidae      | -0.2271 | -0.197  | -0.1273 | -0.0756 | -0.2119 | -0.2629 | 0.1325  |
| 0       | Oligochaeta       | -0.3341 | -0.0235 | -0.161  | 0.2843  | -0.2177 | 0.0139  | -0.1551 |
| 0       | Hydrozoa          | 0.25    | -0.0298 | -0.2017 | 0.0322  | 0.1725  | 0.4043  | 0.0641  |
| 0       | Hirudinea         | 0.055   | 0.0446  | 0.1892  | -0.1354 | 0.0418  | -0.168  | -0.1599 |
| 0       | Acari             | 0.0352  | -0.1392 | 0.0614  | -0.1108 | 0.1744  | -0.0289 | 0.0905  |
| 0       | Ceratopogonidae   | -0.0285 | -0.0617 | 0.0653  | -0.0785 | 0.0732  | -0.0738 | -0.014  |
| 1       | Hexagenia         | 0.0318  | 0.3615  | -0.3629 | -0.5935 | 0.5124  |         |         |
| 1       | Nematoda          | -0.649  | 0.1125  | 0.4104  | 0.0478  | 0.3939  |         |         |
| 1       | Sphaeriidae       | 0.6692  | 0.3492  | 0.4567  | -0.029  | 0.1023  |         |         |
| 1       | Acari             | -0.1957 | 0.5876  | 0.2206  | 0.1236  | -0.4346 |         |         |
| 1       | Hydropsychidae    | 0.0295  | 0.1113  | -0.3554 | 0.6241  | 0.2688  |         |         |
| 1       | Other Trichoptera | 0.0764  | 0.3668  | -0.4262 | 0.3087  | -0.002  |         |         |
| 1       | Dreissena         | -0.0689 | 0.1579  | -0.1118 | -0.1986 | -0.367  |         |         |
| 1       | Caenis            | -0.1821 | 0.0161  | -0.2462 | -0.1038 | -0.245  |         |         |
| 1       | Oligochaeta       | 0.1023  | -0.3702 | -0.0314 | 0.0376  | -0.1142 |         |         |
| 1       | Gastropoda        | -0.0283 | 0.0929  | -0.162  | -0.1604 | -0.1661 |         |         |
| 1       | Chironomidae      | 0.0665  | -0.2309 | -0.0981 | -0.1628 | -0.0419 |         |         |
| 1       | Hydrozoa          | -0.026  | 0.047   | -0.1025 | -0.1049 | -0.2263 |         |         |
| 1       | Ceratopogonidae   | -0.0878 | 0.0088  | -0.0467 | -0.1649 | -0.1398 |         |         |
| 1       | Turbellaria       | -0.087  | 0.0924  | 0.0684  | -0.0459 | 0.0121  |         |         |
| 1       | Hirudinea         | 0.124   | -0.0738 | 0.0844  | 0.0029  | 0.01    |         |         |
| 1       | Amphipoda         | 0.0076  | 0.0452  | -0.0143 | -0.0214 | -0.0505 |         |         |



Table 1.15: Table2 Species Pc Pollution Regression

| Cluster | PC  | Var.(%) | $R^2$  | $\bar{R}^2$ | Coef.   | t-stat | n  |
|---------|-----|---------|--------|-------------|---------|--------|----|
| 0       | PC1 | 23.6    | 0.0751 | 0.0634      | -0.0314 | -2.53  | 81 |
| 0       | PC2 | 13.6    | 0.0667 | 0.0549      | -0.0225 | -2.38  | 81 |
| 0       | PC3 | 10.5    | 0.059  | 0.0471      | -0.0186 | -2.23  | 81 |
| 0       | PC4 | 8.2     | 0      | -0.0127     | 0.0002  | 0.02   | 81 |
| 0       | PC5 | 7       | 0.0287 | 0.0164      | -0.0106 | -1.53  | 81 |
| 0       | PC6 | 6.2     | 0.0005 | -0.0122     | 0.0013  | 0.19   | 81 |
| 0       | PC7 | 6.1     | 0.0186 | 0.0062      | -0.008  | -1.22  | 81 |
| 1       | PC1 | 23.7    | 0.1062 | 0.0636      | -0.0638 | -1.58  | 23 |
| 1       | PC2 | 17.8    | 0.0054 | -0.042      | 0.0124  | 0.34   | 23 |
| 1       | PC3 | 15.4    | 0.1456 | 0.105       | -0.0603 | -1.89  | 23 |
| 1       | PC4 | 12.3    | 0.0196 | -0.0271     | -0.0197 | -0.65  | 23 |
| 1       | PC5 | 7.7     | 0.0062 | -0.0411     | -0.0088 | -0.36  | 23 |

Table 1.16: Table 3a: Pollution PC vs Species PC Regression (Cluster 0)

| Cluster | Poll. PC | Spec. PC | Var.(%) | R <sup>2</sup> | Slope   | n  |
|---------|----------|----------|---------|----------------|---------|----|
| 0       | PC1      | PC1      | 23.6    | 0.0999         | -0.0843 | 81 |
| 0       | PC1      | PC2      | 13.6    | 0.009          | 0.0192  | 81 |
| 0       | PC1      | PC3      | 10.5    | 0.001          | -0.0057 | 81 |
| 0       | PC1      | PC4      | 8.2     | 0.0396         | -0.0312 | 81 |
| 0       | PC1      | PC5      | 7       | 0.006          | -0.0113 | 81 |
| 0       | PC1      | PC6      | 6.2     | 0.0076         | -0.0119 | 81 |
| 0       | PC1      | PC7      | 6.1     | 0.024          | 0.0211  | 81 |
| 0       | PC2      | PC1      | 23.6    | 0.0139         | -0.0288 | 81 |
| 0       | PC2      | PC2      | 13.6    | 0.0818         | -0.053  | 81 |
| 0       | PC2      | PC3      | 10.5    | 0.0195         | 0.0227  | 81 |
| 0       | PC2      | PC4      | 8.2     | 0.0271         | -0.0236 | 81 |
| 0       | PC2      | PC5      | 7       | 0.0028         | -0.0071 | 81 |
| 0       | PC2      | PC6      | 6.2     | 0.0075         | -0.0108 | 81 |
| 0       | PC2      | PC7      | 6.1     | 0.0375         | -0.024  | 81 |
| 0       | PC3      | PC1      | 23.6    | 0.0407         | -0.0488 | 81 |
| 0       | PC3      | PC2      | 13.6    | 0.0038         | -0.0114 | 81 |
| 0       | PC3      | PC3      | 10.5    | 0.0774         | -0.045  | 81 |
| 0       | PC3      | PC4      | 8.2     | 0.0438         | 0.0298  | 81 |
| 0       | PC3      | PC5      | 7       | 0.009          | -0.0125 | 81 |
| 0       | PC3      | PC6      | 6.2     | 0.0017         | 0.0051  | 81 |
| 0       | PC3      | PC7      | 6.1     | 0.0026         | -0.0063 | 81 |
| 0       | PC4      | PC1      | 23.6    | 0.0029         | 0.0138  | 81 |
| 0       | PC4      | PC2      | 13.6    | 0.0017         | -0.008  | 81 |
| 0       | PC4      | PC3      | 10.5    | 0              | 0.0004  | 81 |
| 0       | PC4      | PC4      | 8.2     | 0.0057         | -0.0114 | 81 |
| 0       | PC4      | PC5      | 7       | 0.0078         | -0.0124 | 81 |
| 0       | PC4      | PC6      | 6.2     | 0.0012         | -0.0045 | 81 |
| 0       | PC4      | PC7      | 6.1     | 0.0026         | -0.0067 | 81 |
| 0       | PC5      | PC1      | 23.6    | 0.0034         | -0.014  | 81 |
| 0       | PC5      | PC2      | 13.6    | 0.0518         | -0.0417 | 81 |
| 0       | PC5      | PC3      | 10.5    | 0.0072         | -0.0136 | 81 |
| 0       | PC5      | PC4      | 8.2     | 0.016          | 0.018   | 81 |
| 0       | PC5      | PC5      | 7       | 0.0056         | -0.0098 | 81 |
| 0       | PC5      | PC6      | 6.2     | 0.0007         | -0.0033 | 81 |
| 0       | PC5      | PC7      | 6.1     | 0.0036         | -0.0074 | 81 |
| 0       | PC6      | PC1      | 23.6    | 0.008          | 0.0224  | 81 |
| 0       | PC6      | PC2      | 13.6    | 0.0034         | -0.0112 | 81 |
| 0       | PC6      | PC3      | 10.5    | 0.0722         | -0.045  | 81 |
| 0       | PC6      | PC4      | 8.2     | 0.0002         | 0.002   | 81 |
| 0       | PC6      | PC5      | 7       | 0.0045         | -0.0091 | 81 |
| 0       | PC6      | PC6      | 6.2     | 0.0414         | 0.0262  | 81 |
| 0       | PC6      | PC7      | 6.1     | 0.0166         | -0.0164 | 81 |

Table 1.17: Table 3b: Pollution PC vs Species PC Regression (Cluster 1)

| Cluster | Poll. PC | Spec. PC | Var.(%) | R <sup>2</sup> | Slope   | n  |
|---------|----------|----------|---------|----------------|---------|----|
| 1       | PC1      | PC1      | 23.7    | 0.0484         | -0.0607 | 23 |
| 1       | PC1      | PC2      | 17.8    | 0              | -0.0007 | 23 |
| 1       | PC1      | PC3      | 15.4    | 0.0332         | -0.0406 | 23 |
| 1       | PC1      | PC4      | 12.3    | 0.0002         | -0.0026 | 23 |
| 1       | PC1      | PC5      | 7.7     | 0.1592         | 0.063   | 23 |
| 1       | PC2      | PC1      | 23.7    | 0.0849         | -0.1146 | 23 |
| 1       | PC2      | PC2      | 17.8    | 0.0077         | 0.03    | 23 |
| 1       | PC2      | PC3      | 15.4    | 0.1271         | -0.1132 | 23 |
| 1       | PC2      | PC4      | 12.3    | 0.0631         | -0.0711 | 23 |
| 1       | PC2      | PC5      | 7.7     | 0.0526         | -0.0516 | 23 |
| 1       | PC3      | PC1      | 23.7    | 0.0593         | 0.1086  | 23 |
| 1       | PC3      | PC2      | 17.8    | 0.0004         | 0.0078  | 23 |
| 1       | PC3      | PC3      | 15.4    | 0.0043         | -0.0235 | 23 |
| 1       | PC3      | PC4      | 12.3    | 0.0211         | -0.0466 | 23 |
| 1       | PC3      | PC5      | 7.7     | 0.251          | -0.1279 | 23 |
| 1       | PC4      | PC1      | 23.7    | 0.0608         | -0.0607 | 23 |
| 1       | PC4      | PC2      | 17.8    | 0.08           | 0.0603  | 23 |
| 1       | PC4      | PC3      | 15.4    | 0.3287         | -0.1139 | 23 |
| 1       | PC4      | PC4      | 12.3    | 0.0661         | -0.0455 | 23 |
| 1       | PC4      | PC5      | 7.7     | 0              | -0.0003 | 23 |
| 1       | PC5      | PC1      | 23.7    | 0.1568         | -0.1837 | 23 |
| 1       | PC5      | PC2      | 17.8    | 0.0025         | -0.0203 | 23 |
| 1       | PC5      | PC3      | 15.4    | 0.0402         | 0.0751  | 23 |
| 1       | PC5      | PC4      | 12.3    | 0.0238         | 0.0515  | 23 |
| 1       | PC5      | PC5      | 7.7     | 0.0482         | -0.0583 | 23 |
| 1       | PC6      | PC1      | 23.7    | 0.0028         | 0.0157  | 23 |
| 1       | PC6      | PC2      | 17.8    | 0.0044         | 0.017   | 23 |
| 1       | PC6      | PC3      | 15.4    | 0.0391         | -0.0471 | 23 |
| 1       | PC6      | PC4      | 12.3    | 0.0001         | -0.0022 | 23 |
| 1       | PC6      | PC5      | 7.7     | 0.0096         | 0.0165  | 23 |

### 1.3. 03 Env-Driven Taxa clusters (LDA)

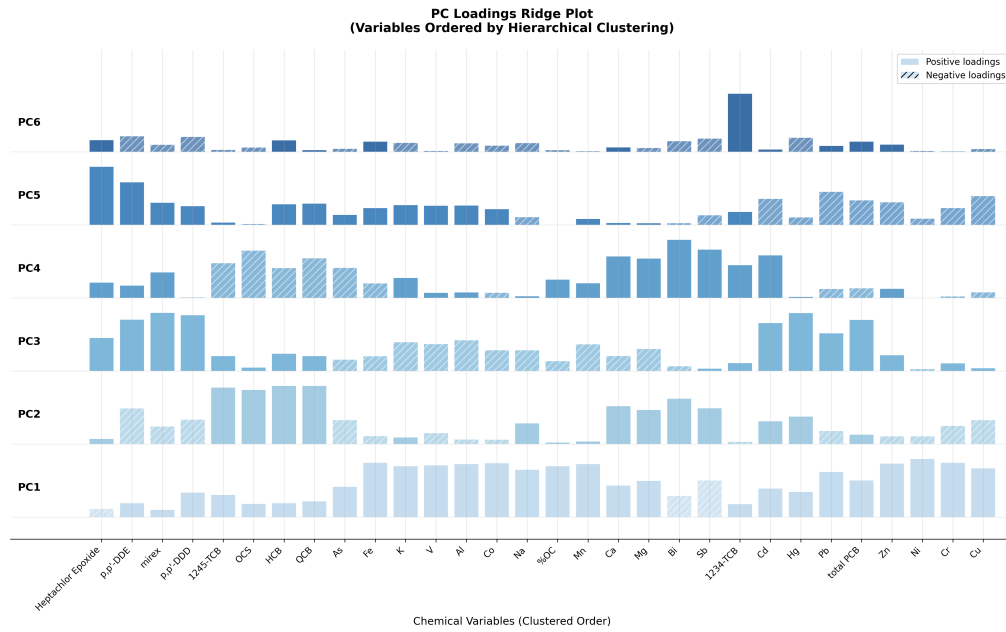


Figure 1.1: PC loadings ridge plot

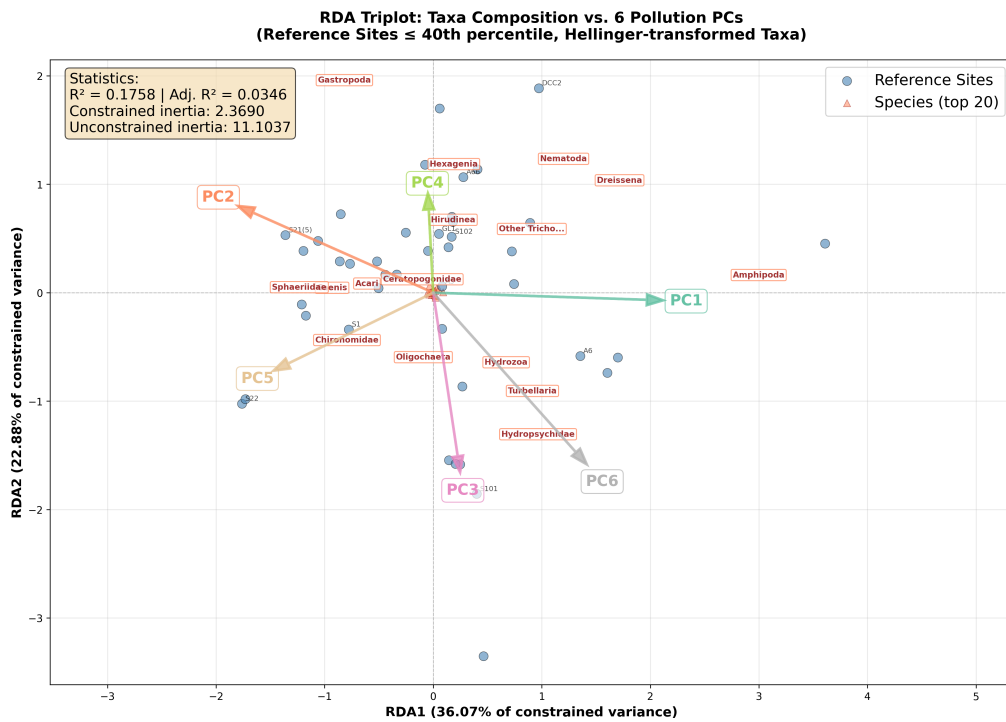


Figure 1.2: RDA validation plot of Pollution PCs vs Species Matrix

### 1.3. 03 Env-Driven Taxa clusters (LDA)

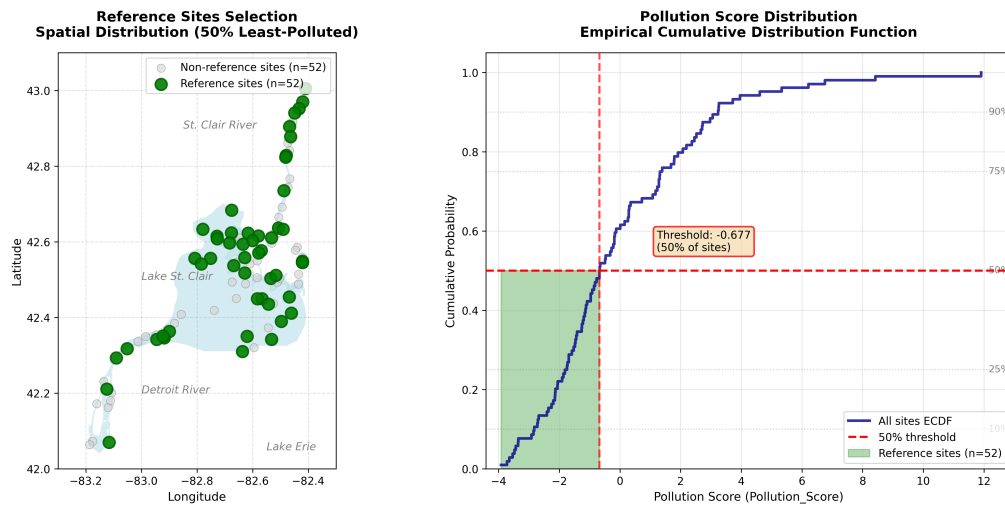
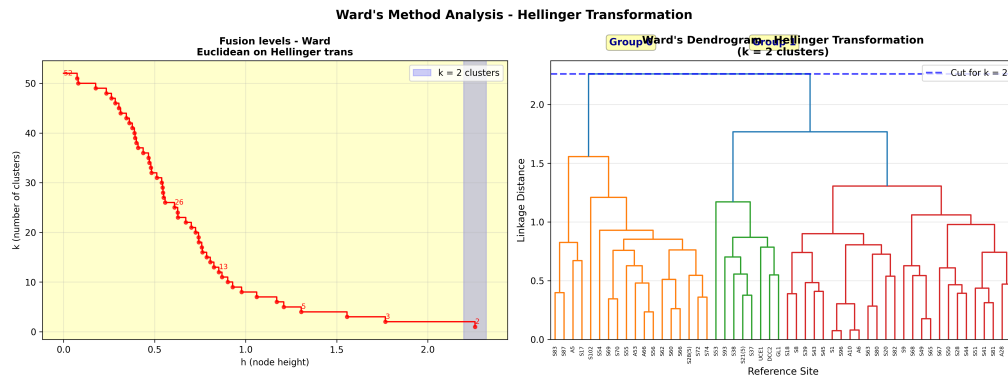
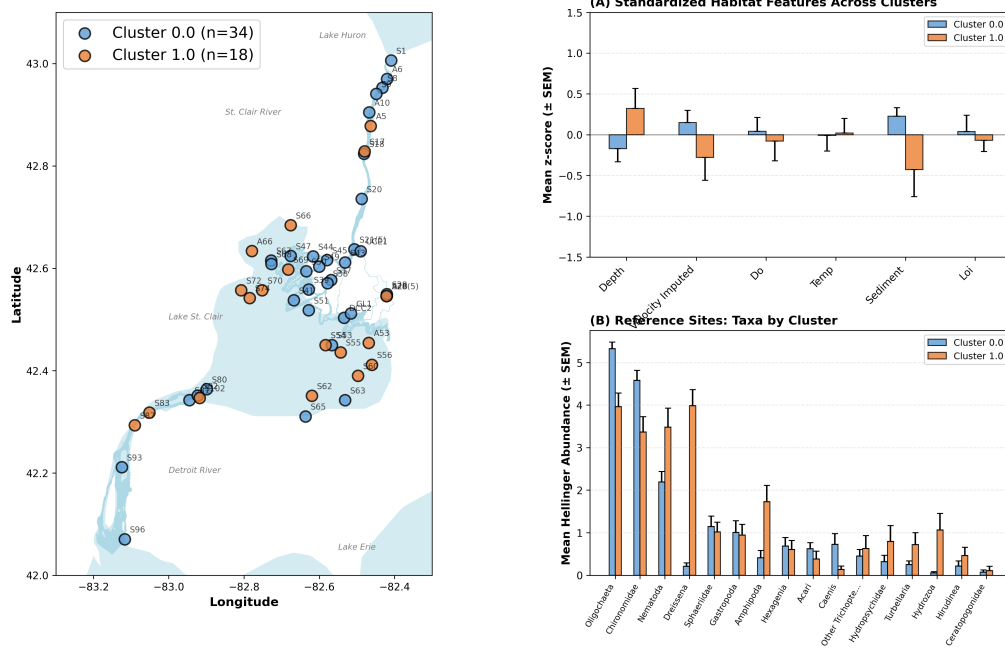


Figure 1.3: Reference site selection based on  $p\%$  lowest pollution scores



#### Reference Sites Cluster Analysis - Geographic, Environmental & Taxa Patterns



### 1.3. 03 Env-Driven Taxa clusters (LDA)

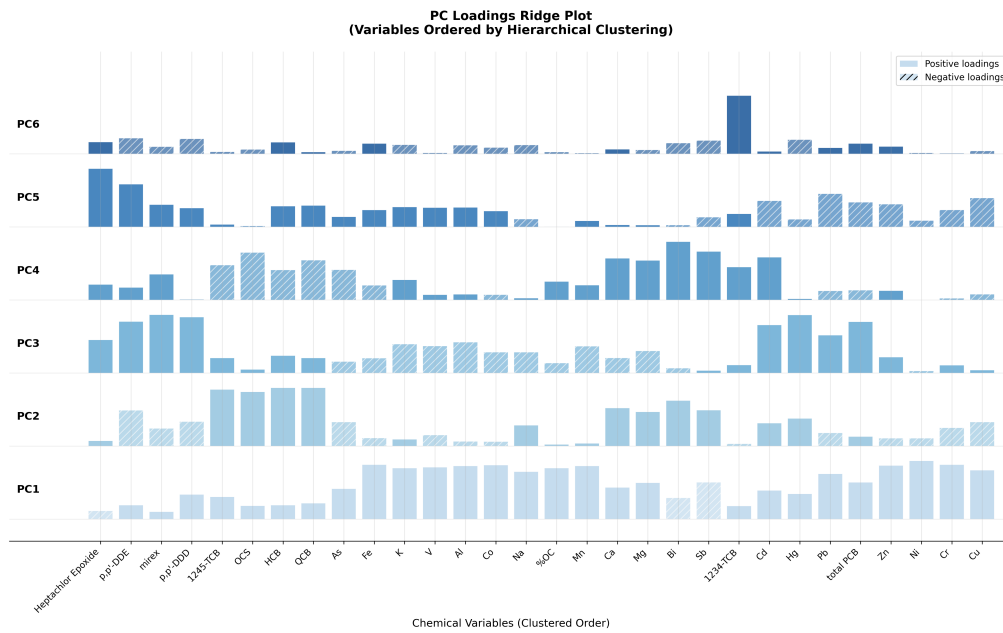


Figure 1.4: PC loadings ridge plot

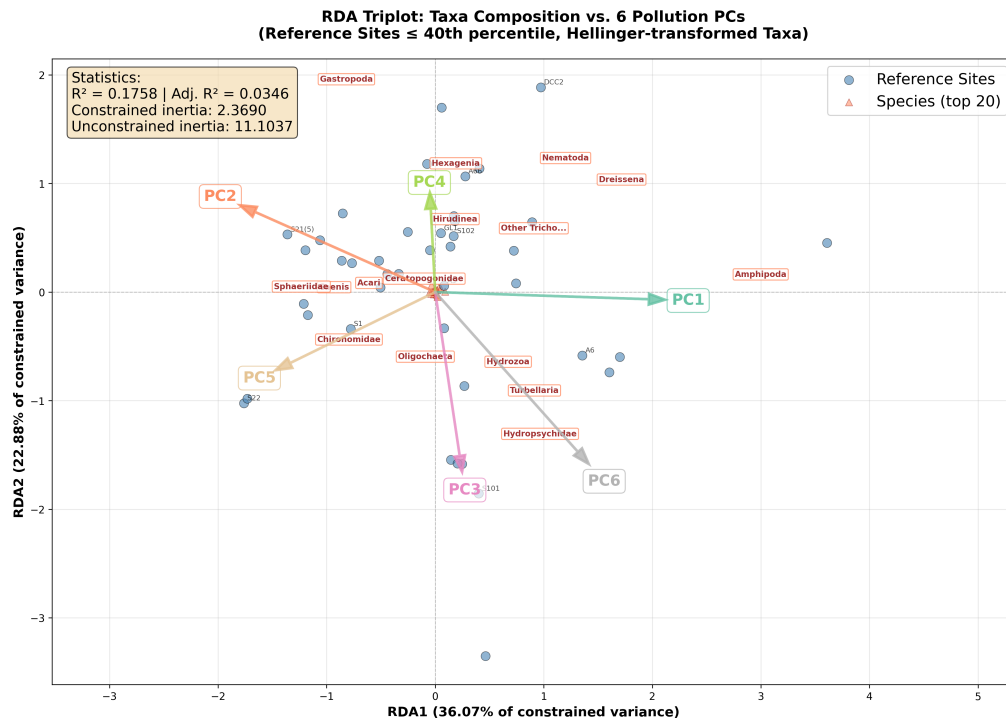


Figure 1.5: RDA validation plot of Pollution PCs vs Species Matrix

### 1.3. 03 Env-Driven Taxa clusters (LDA)

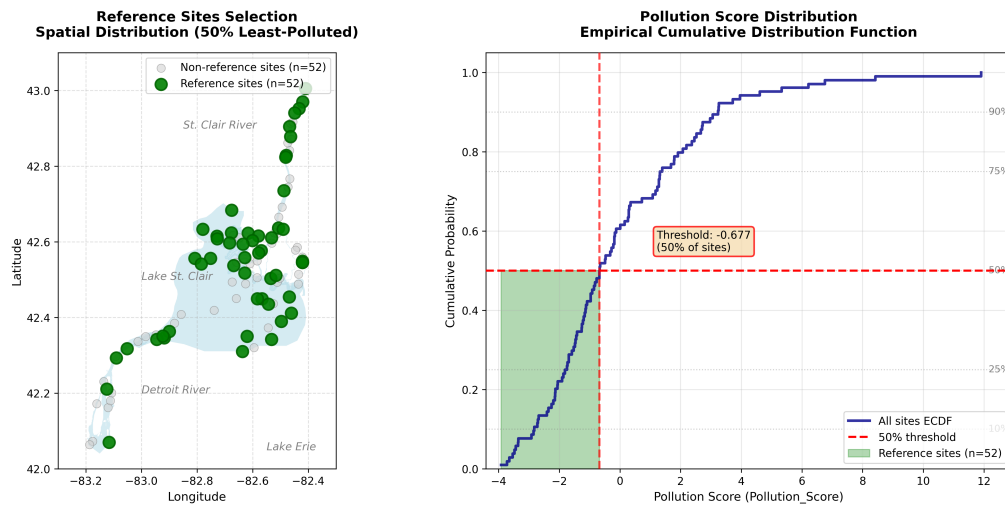
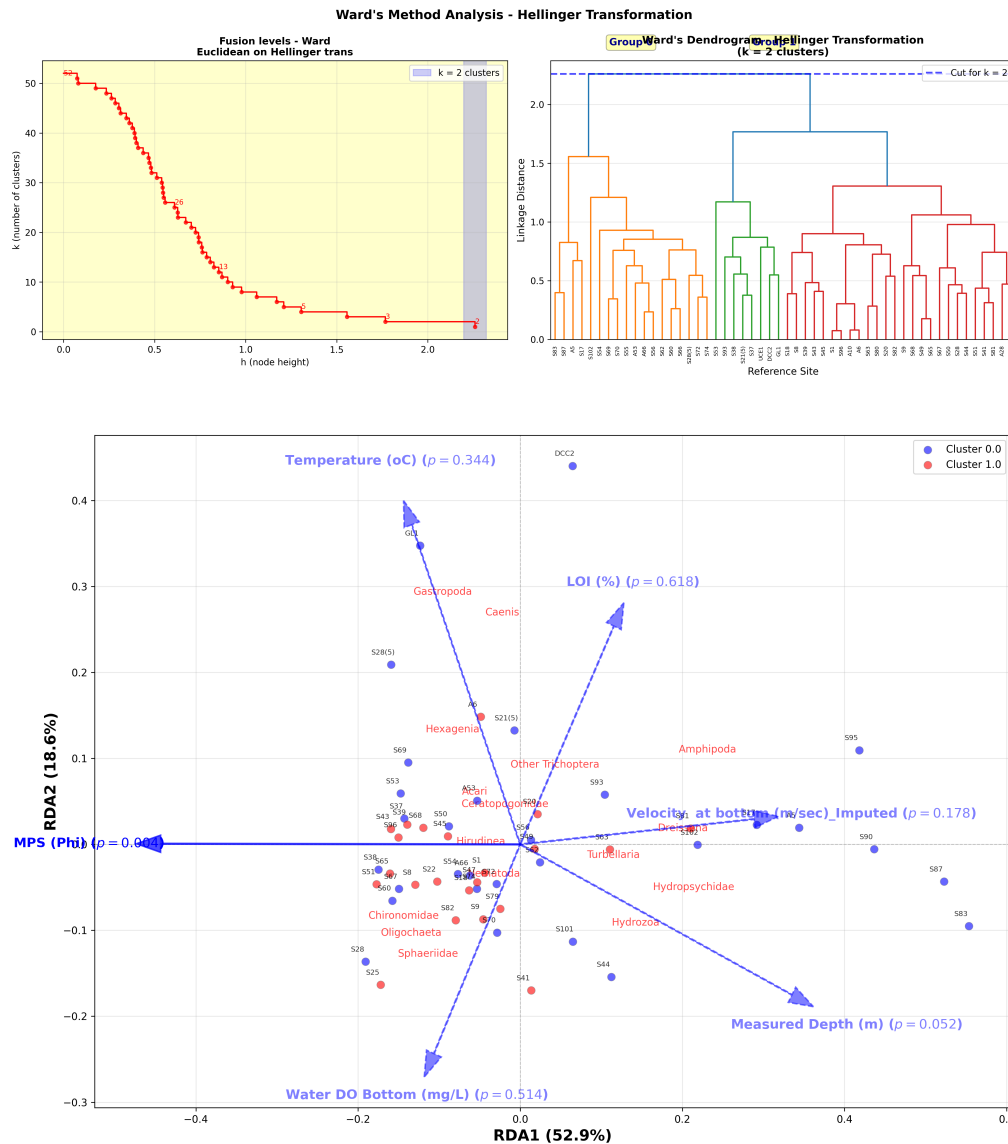
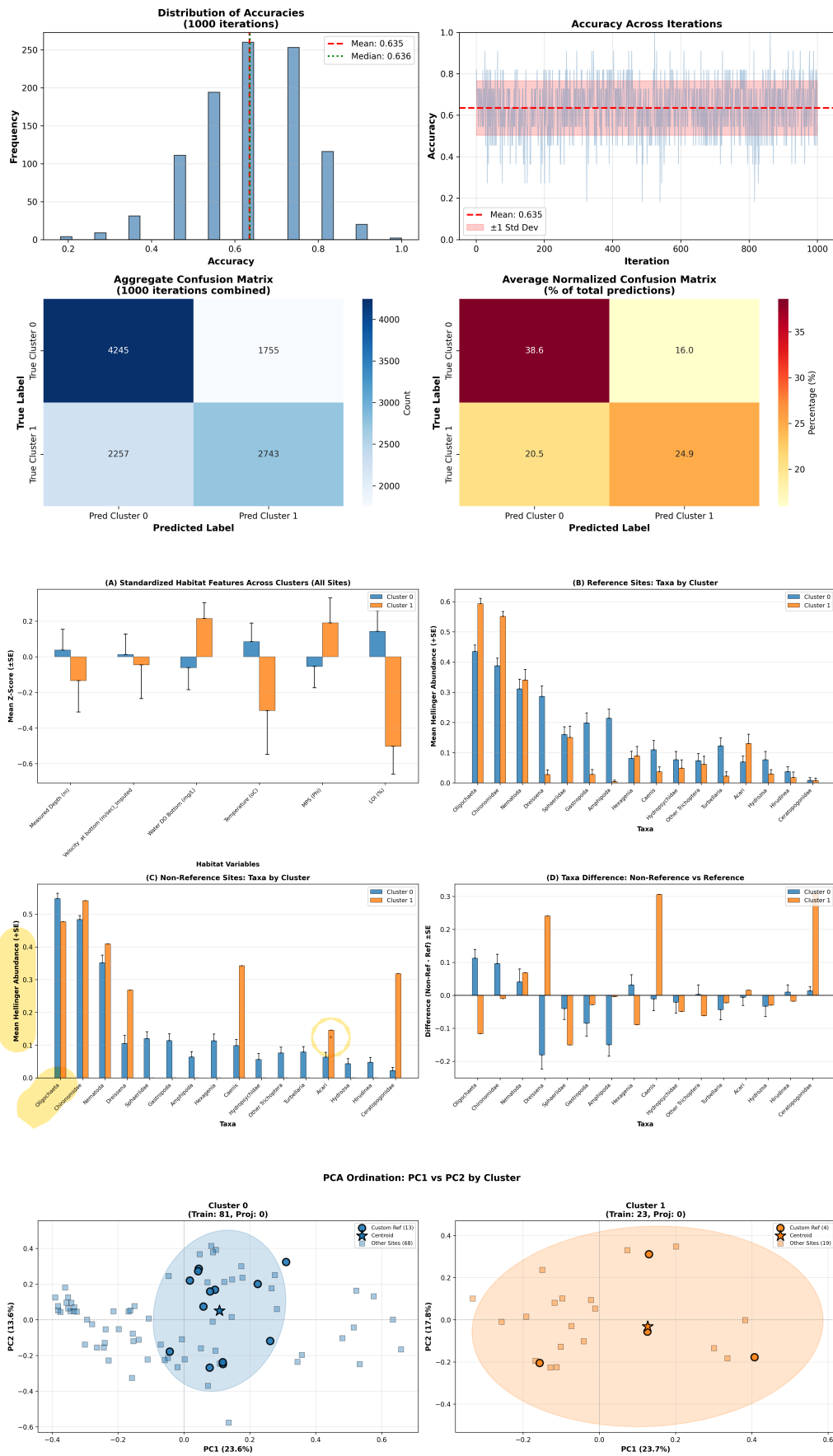


Figure 1.6: Reference site selection based on  $p\%$  lowest pollution scores



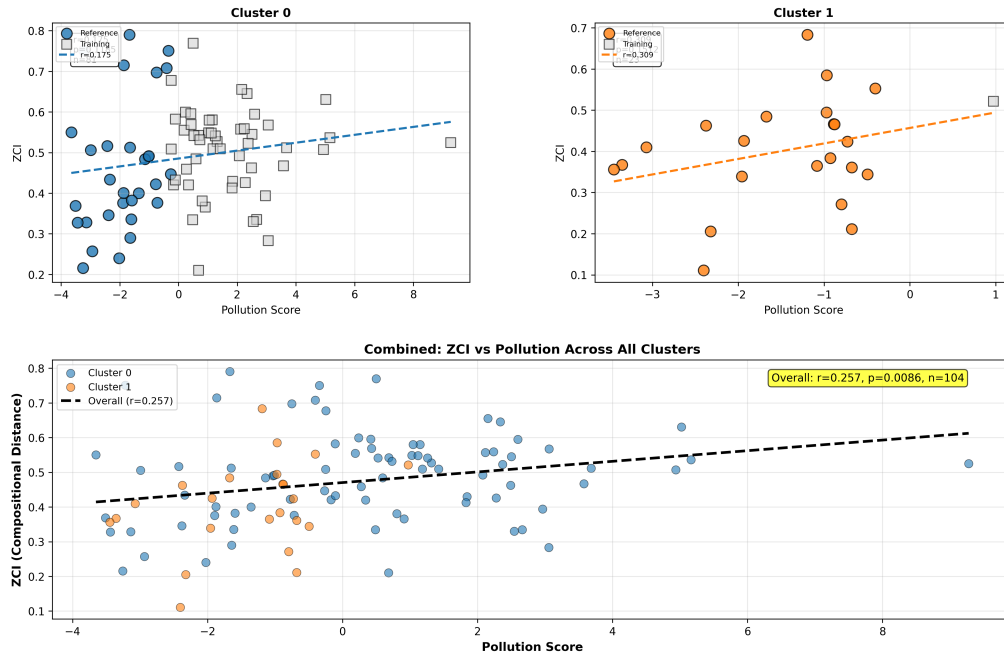
### 1.3. 03 Env-Driven Taxa clusters (LDA)



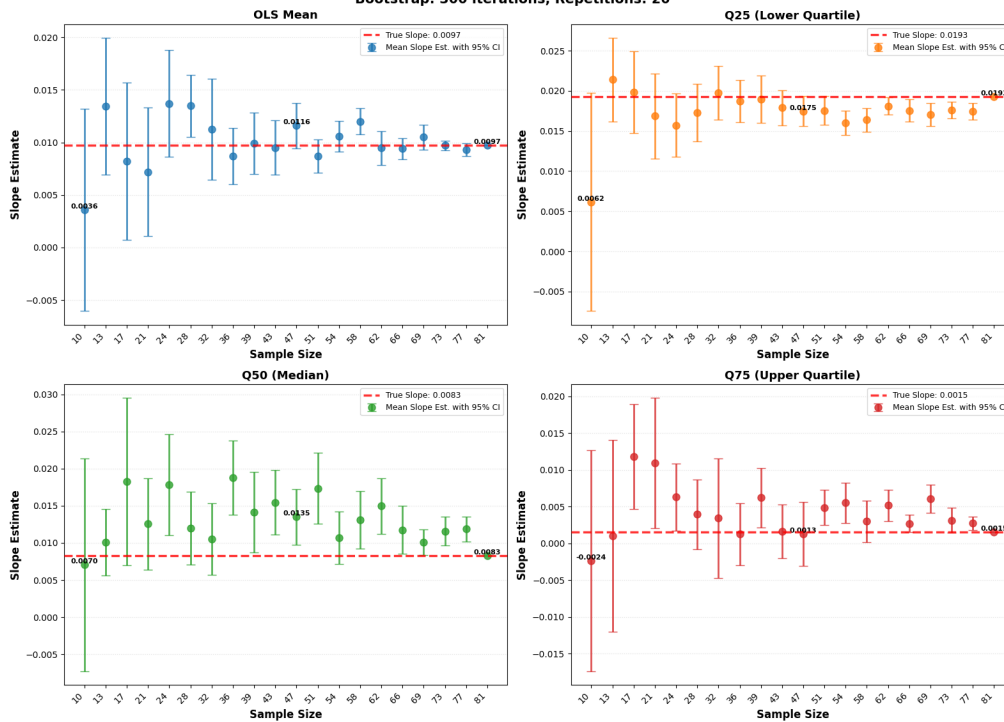


### 1.3. 03 Env-Driven Taxa clusters (LDA)

**Zoobenthic Community Indicator (ZCI) vs Pollution Score**



**Cluster 0: 95% CI Coverage vs Sample Size**  
Bootstrap: 300 iterations, Repetitions: 20



### 1.3. 03 Env-Driven Taxa clusters (LDA)

